

Euclidean Geometry In Mathematical Olympiads

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Problem 1.36



Chapter 1

fgm

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similar sol, just for practice



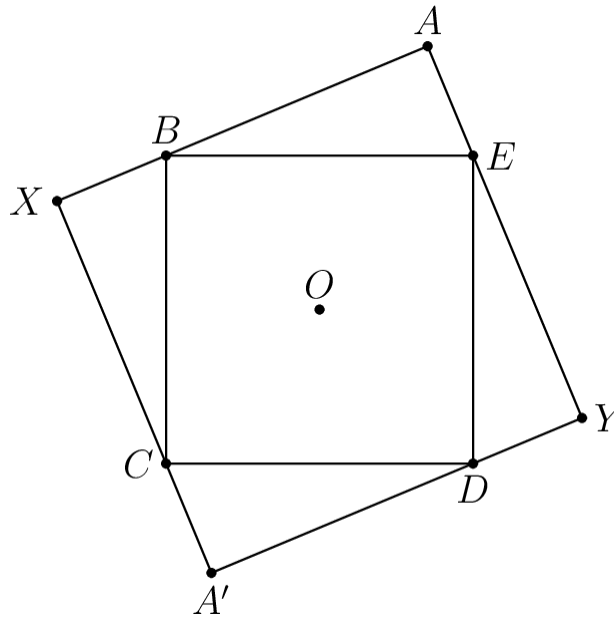
Let P be the midpoint of BC . We note that $BP = EP = OP$, since O is symmetrical about the four vertices, hence O also lies on the circle with diameter BE and A on it. In $\triangle BOE$ by symmetry, we have $\angle OBE = \angle BEO = 45^\circ$. By simple angle chasing in the cyclic quadrilateral $ABOE$, we get $\angle BAO = \angle OAE = 45^\circ$. Thus AO is bisecting $\angle BAE$. ■

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Reflect A over O and construct a square:The squares are concentric and clearly AA' bisects $\angle XAY$.

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Alternatively, we can bary bash.

Rename $E \mapsto C$ and set $A = (1, 0, 0), B = (0, 1, 0), C = (0, 0, 1)$. By Conway's